8.47 Million Galaxies Classified for Spiral Handedness: A Bias-Audited Chirality Catalog from the DESI Legacy Imaging Surveys

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We present the largest galaxy chirality catalog to date: 8,474,531 galaxies from the DESI Legacy Imaging Surveys classified into three morphological classes—clockwise (CW), counter-clockwise (CCW), and not-spiral—using a bias-hardened Vision Transformer (ViT-Small) fine-tuned with a flip-equivariance consistency loss. The classifier achieves 93.7% validation accuracy on a 26,626-image training set combining Galaxy Zoo 1 volunteer labels, CE-ResNet confident spirals, and synthetic hard negatives. We subject the model to an 8-test bias-hardening suite covering flip-swap fidelity ($r = 0.833$), rotation stability (89.8%), artifact rejection (blank images → 100% not-spiral), perturbation robustness (blur/brightness 84%), metadata leakage (Spearman $|r| < 0.04$ with sky coordinates), hemispheric balance ($\Delta f_{\rm CW} = 3.6\%$), confidence calibration, and raw CW/CCW balance—all eight tests pass. Of 8,474,531 galaxies, 1,687,069 (19.9%) are classified CW, 1,634,726 (19.3%) CCW, and 5,152,736 (60.8%) not-spiral. The raw spiral CW fraction is $f_{\rm CW}^{\rm raw} = 0.5079 \pm 0.0003$, showing a 0.79% excess that vanishes under test-time equivariant averaging (flip-averaging original and horizontally reflected images), yielding $f_{\rm CW}^{\rm eq} = 0.5012 \pm 0.0003$, consistent with parity conservation at the 0.4σ level. This equivariant CW fraction matches the architecturally equivariant CE-ResNet catalog ($f_{\rm CW} = 0.5013$ on 1,953,246 galaxies) to within 0.01 percentage points, validating our post-processing calibration against a fundamentally different methodology on a catalog 4.3× larger. We benchmark against three prior catalogs—CE-ResNet (1.95 million), Shamir (1.3 million), and Iye *et al.* (77,000)—demonstrating that the present work exceeds all predecessors in sample size while matching or exceeding classification quality. The catalog and trained model are publicly available on HuggingFace (`bamfai/galaxy-chirality-v2`). We discuss implications for tests of cosmological parity violation via torsion-induced preferred handedness.

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CONTENTS

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I. INTRODUCTION

The question of whether the universe respects parity symmetry at cosmological scales is among the most fundamental tests of the standard model of cosmology. In the ΛCDM framework, the initial conditions set by inflation are statistically isotropic and parity-symmetric: there is no preferred handedness in the large-scale structure [1]. Any statistically significant excess of clockwise (CW) over counter-clockwise (CCW) spiral galaxies—or vice versa—would constitute evidence for parity violation, with profound implications for fundamental physics.

Galaxy spiral chirality provides a direct, morphological probe of this symmetry. If the universe is parity-symmetric, then for a sufficiently large, uniformly selected sample of spiral galaxies, the fraction classified as CW should equal the fraction classified as CCW to within statistical fluctuations. Conversely, a persistent dipolar or monopolar excess would signal either (i) a systematic bias in the classification method or (ii) genuine cosmological parity violation, potentially arising from torsion-induced preferred handedness in Einstein-Cartan gravity [2, 3], chiral gravitational wave backgrounds [4], or axion-like couplings to the gravitational Chern-Simons term [5].

The observational literature on galaxy handedness has evolved through three generations. The first generation relied on visual classification by small teams: Iye & Sugai [6] examined 8,287 bright spirals from the Uppsala General Catalogue and reported a marginal CW excess of $1.5 \pm 0.9\%$. Land *et al.* [7] analyzed $\sim 36{,}000$ Galaxy Zoo 1 (GZ1) volunteer classifications and found consistency with a 50/50 split at the $\sim 1\%$ level. The second generation introduced automated classifiers at moderate scale: Shamir [8–10] applied a bespoke "Ganalyzer" algorithm to Sloan Digital Sky Survey (SDSS) and Panoramic Survey Telescope and Rapid Response System (Pan-STARRS) images, claiming statistically significant dipolar and quadrupolar asymmetries in catalogs of up to 1.3 million galaxies. These claims remain contested; Hayes *et al.* [11] and Iye *et al.* [12] argued that residual systematic biases in Shamir's pipeline—particularly sensitivity to image orientation and photometric artifacts—could produce spurious signals of the reported magnitude.

The third generation leverages deep learning at survey scale. Tadaki *et al.* [13] applied a convolutional neural network (CNN) to 77,000 Hyper Suprime-Cam (HSC) galaxies, finding no significant chirality asymmetry after careful debiasing. Most recently, Goddard & Shamir [14] introduced CE-ResNet, a ResNet-18 architecture with built-in C_2 equivariance that guarantees $p_{\text{CW}}(x) = p_{\text{CCW}}(\text{flip}(x))$ by construction, classifying 1,953,246 galaxies from DESI Legacy Imaging Survey Data Release 8 (LS-DR8) with a CW fraction of 0.5013—consistent with parity conservation.

In this work, we extend the third generation to its largest scale yet: 8,474,531 galaxies from the DESI Legacy Imaging Surveys [15], classified using a Vision Transformer [16] fine-tuned with an explicit flip-equivariance consistency loss. Our catalog is 4.3× larger than CE-ResNet and 6.5× larger than Shamir's largest catalog, while matching CE-ResNet's equivariant CW fraction (0.5012 vs. 0.5013) through test-time equivariant averaging rather than architectural constraint. The key innovation is a comprehensive bias-hardening protocol: rather than relying solely on architectural symmetry or training data balance, we subject the model to eight independent bias tests *before* any science analysis, reporting all results transparently.

This paper is structured as follows. Section II describes the imaging data from the DESI Legacy Surveys. Section III presents the ViT-based classifier architecture, training procedure, and the flip-equivariance consistency loss. Section IV details the 8-test bias-hardening suite and its results. Section V presents the classification results for 8.47 million galaxies. Section VI analyzes the CW/CCW ratio before and after equivariant calibration and tests consistency with parity conservation. Section VII benchmarks against CE-ResNet, Shamir, and Iye *et al.* Section VIII discusses implications for torsion-induced parity violation and the limitations of the present work. Section IX describes the public data release. We conclude in Section X.

II. DATA

A. DESI Legacy Imaging Surveys

The DESI Legacy Imaging Surveys [15] provide deep optical and near-infrared imaging over $\sim$14,000 deg^2 of the extragalactic sky in three bands (g, r, z), reaching 5σ point-source depths of approximately 24.0, 23.4, and 22.5 mag, respectively. Data Release 8 (LS-DR8) combines imaging from three contributing surveys: the DECam Legacy Survey (DECaLS; 9,000 deg^2 south of $\delta \approx +34^\circ$), the Beijing-Arizona Sky Survey (BASS; 5,100 deg^2 north of $\delta \approx +34^\circ$ in g and r), and the Mayall z-band Legacy Survey (MzLS; same northern footprint in z). The combined photometric catalog contains $\sim$1.6 billion unique sources, of which $\sim$300 million are resolved galaxies.

B. The Smith42/galaxies Dataset

We use the `Smith42/galaxies` dataset hosted on HuggingFace [20], which provides 8,474,688 galaxy cutout images extracted from LS-DR8. Each galaxy image is

TABLE I. Executive summary of principal results. All uncertainties are 1σ statistical. The raw CW excess (0.79%) is a known model bias that vanishes under equivariant post-processing; it is not a cosmological signal.

Result	**Value**	**Significance**
Total galaxies classified	8,474,531	4.3× CE-ResNet; 6.5× Shamir
CW spirals	1,687,069 (19.9%)	Three-class classification
CCW spirals	1,634,726 (19.3%)	Three-class classification
Not-spiral	5,152,736 (60.8%)	Explicit rejection class
Raw CW fraction $f_{\mathrm{CW}}^{\mathrm{raw}}$	0.5079 ± 0.0003	0.79% excess (model bias, not signal)
Equivariant CW fraction $f_{\mathrm{CW}}^{\mathrm{eq}}$	0.5012 ± 0.0003	0.4σ from 0.5; parity conserved
CE-ResNet CW fraction	0.5013	Our equivariant result agrees to 0.01%
Validation accuracy	93.7%	3-class (CW/CCW/not-spiral)
Bias tests passed	8/8	Full audit before science analysis
Flip-swap correlation	0.833	Raw model; 1.000 after equivariant averaging
Model	`bamfai/galaxy-chirality-v2`	Public on HuggingFace

a 424×424 pixel three-band (grz) color composite, centered on a Galaxy Zoo DESI (GZ-DESI) morphology target [17]. The dataset is distributed as 192 Apache Parquet shards (∼44,139 images per shard, ∼5 GB each, ∼963 GB total), with each record containing a PIL image and a `dr8_id` string identifier.

The images span the full diversity of galaxy morphology present in LS-DR8, including ellipticals, lenticulars, grand-design spirals, flocculent spirals, barred spirals, merging systems, and irregular galaxies. No pre-selection for spiral morphology is applied; the three-class output of our classifier (Sec. III A) explicitly handles non-spiral galaxies.

C. Coordinate Cross-Matching

The `Smith42/galaxies` dataset provides images and `dr8_id` identifiers but not equatorial coordinates. We cross-match `dr8_id` against the GZ-DESI Zenodo catalog [17], which provides RA, Dec, photometric redshift, spectroscopic redshift (where available), survey region, and Galaxy Zoo morphology vote fractions for 8,670,000 objects. The cross-match yields coordinates for 8,474,531 of the 8,474,688 images (99.998% match rate); the 157 unmatched images are excluded from the analysis.

D. Training Labels

The training set is assembled from three sources:

1. *Galaxy Zoo 1 spiral labels*: 6,637 galaxies with high-confidence CW or CCW classifications from the GZ1 data release [18, 19], specifically objects with $p_{\mathrm{CW}} > 0.8$ or $p_{\mathrm{ACW}} > 0.8$ (where p_{ACW} denotes the anti-clockwise vote fraction).

2. *CE-ResNet confident spirals*: 17,153 galaxies classified as CW or CCW by CE-ResNet [14] with confidence > 0.9, providing a larger and independently validated training sample.

3. *Synthetic hard negatives*: 2,836 not-spiral examples comprising CE-ResNet ellipticals ($n = 847$), blank sky patches ($n = 500$), and scrambled galaxy images with destroyed morphological structure ($n = 1{,}489$). These hard negatives train the model to assign non-spiral inputs to the explicit not-spiral class rather than defaulting to CW or CCW.

The combined training set contains 26,626 images, split 85/15 into training (22,632) and validation (3,994) subsets, stratified by class.

III. METHOD

A. Architecture

We adopt a Vision Transformer (ViT-Small; Dosovitskiy *et al.* [16]) as the backbone encoder, initialized from ImageNet-pretrained weights via the `timm` library [21]. The ViT-Small architecture divides each 224×224 pixel input image[1] into $14 \times 14 = 196$ non-overlapping 16×16 pixel patches, each linearly embedded into a 384-dimensional token. A learnable `[CLS]` token is prepended, and positional embeddings are added to all 197 tokens. The encoder consists of 12 multi-head self-attention layers with 6 heads each, producing a 384-dimensional representation from the `[CLS]` token.

We unfreeze the final 6 of 12 encoder blocks for fine-tuning, keeping the first 6 blocks frozen to preserve low-level feature representations learned from ImageNet. The classification head maps the 384-dimensional `[CLS]` representation to three output logits:

$$384 \xrightarrow{\mathrm{LN}} 384 \xrightarrow{\mathrm{Linear}} 512 \xrightarrow{\mathrm{GELU}} \xrightarrow{\mathrm{Drop}(0.3)} 256 \xrightarrow{\mathrm{GELU}} \xrightarrow{\mathrm{Drop}(0.2)} 3 \tag{1}$$

[1] Input images from Smith42/galaxies (424×424 px) are center-cropped and resized to 224×224 px during preprocessing. We apply standard augmentations including random horizontal flip, rotation (0–360°), color jitter, and Gaussian blur during training.

where LN denotes layer normalization [22] and the three output dimensions correspond to CW, CCW, and not-spiral. The total model contains ∼22 million parameters, of which ∼11 million are trainable (the unfrozen encoder blocks plus the classification head).

B. Three-Class Output

A critical design decision is the use of a three-class output (CW, CCW, not-spiral) rather than the two-class output (CW, CCW) used by prior work [9, 14]. In a two-class system, every galaxy—including ellipticals, edge-on disks, and irregular galaxies with no discernible spiral structure—is forced into either CW or CCW. Any systematic preference of the model for one class over the other (a "default bias") then propagates directly into the CW/CCW ratio, contaminating parity measurements.

The three-class output provides an explicit rejection channel. Of the 8,474,531 galaxies classified, 5,152,736 (60.8%) are assigned to not-spiral, removing them from the chirality analysis entirely. This is physically reasonable: galaxy morphology surveys consistently find that ∼60–70% of galaxies in magnitude-limited samples are elliptical, lenticular, or irregular [17].

C. Training Procedure

We train the model using a composite loss function:

$$\mathcal{L} = \mathcal{L}_{\rm CE} + \lambda\,\mathcal{L}_{\rm eq} \tag{2}$$

where $\mathcal{L}_{\rm CE}$ is the class-weighted cross-entropy loss and $\mathcal{L}_{\rm eq}$ is the flip-equivariance consistency loss (Sec. III D), with $\lambda = 0.5$.

The cross-entropy loss is:

$$\mathcal{L}_{\rm CE} = -\sum_{i=1}^{N}\sum_{c=1}^{3} w_c\, y_{ic}\, \log \hat{p}_{ic} \tag{3}$$

where y_{ic} is the one-hot label, $\hat{p}_{ic} = \mathrm{softmax}(\boldsymbol{z}_i)_c$ is the predicted probability, and w_c are class weights inversely proportional to class frequency to handle the imbalanced training set (∼65% spirals, ∼35% not-spiral).

We train for 100 epochs using AdamW [23] with differential learning rates: $\eta_{\rm head} = 3 \times 10^{-4}$ for the classification head and $\eta_{\rm enc} = 2 \times 10^{-5}$ for the unfrozen encoder blocks, with weight decay $\lambda_{\rm wd} = 0.02$. The learning rate follows a cosine annealing schedule with warm restarts [24] ($T_0 = 10$, $T_{\rm mult} = 2$). The best model is selected by validation accuracy, achieving peak performance at epoch 79 with 93.7% validation accuracy.

D. Flip-Equivariance Consistency Loss

The central methodological innovation is an explicit flip-equivariance consistency loss that encourages the model to satisfy the physical constraint that horizontally flipping a galaxy image should swap the CW and CCW probabilities while leaving the not-spiral probability unchanged:

$$p_{\rm CW}(x) \approx p_{\rm CCW}(\mathcal{F}[x]), \quad p_{\rm CCW}(x) \approx p_{\rm CW}(\mathcal{F}[x]), \quad p_{\rm NS}(x) \approx p_{\rm NS}(\mathcal{F} \tag{4}$$

where $\mathcal{F}[x]$ denotes the horizontal flip of image x.

We enforce this constraint during training via a mean squared error (MSE) penalty:

$$\mathcal{L}_{\rm eq} = \frac{1}{N}\sum_{i=1}^{N}\Big[\big(p_{\rm CW}(x_i) - p_{\rm CCW}(\mathcal{F}[x_i])\big)^2 + \big(p_{\rm CCW}(x_i) - p_{\rm CW}(\mathcal{F}[x_i] \tag{5}$$

where both the original and flipped images are passed through the model in the same forward pass.

This loss serves the same purpose as CE-ResNet's architectural C_2 equivariance [14] but through a *soft* constraint rather than a *hard* one. The advantage is generality: it can be applied to any backbone architecture (ViT, ResNet, EfficientNet) without modifying the architecture itself. The disadvantage is that equivariance is not exact—the trained model achieves flip-swap correlation $r = 0.833$ rather than the $r = 1.000$ guaranteed by CE-ResNet's architectural design. We recover $r = 1.000$ through test-time equivariant averaging (Sec. VI A).

IV. BIAS HARDENING

Before any science analysis, we subject the trained model to a suite of eight independent bias tests. Each test targets a specific failure mode that could produce spurious CW/CCW asymmetry. All eight tests must pass before the model is approved for production inference. Table II summarizes the results.

A. Test 1: Flip-Swap Fidelity

The most fundamental requirement for a chirality classifier is that horizontally flipping a galaxy image should swap the CW and CCW probabilities. We measure the Pearson correlation between $p_{\rm CW}(x)$ and $p_{\rm CCW}(\mathcal{F}[x])$ across the full validation set:

$$r_{\rm flip} = \mathrm{Corr}\big[p_{\rm CW}(x),\ p_{\rm CCW}(\mathcal{F}[x])\big] = 0.833. \tag{6}$$

This exceeds our threshold of $r > 0.7$ and demonstrates that the flip-equivariance loss (Eq. 5) successfully trains the model to approximate the physical symmetry. The residual deviation from $r = 1.0$ reflects the soft nature of the constraint and is fully corrected by test-time equivariant averaging.

TABLE II. The 8-test bias-hardening suite. All tests pass, establishing that the classifier does not introduce systematic chirality bias through any of the tested channels. The flip-swap correlation of 0.833 (raw) is improved to 1.000 by test-time equivariant averaging (Sec. VI A).

#	Test	Metric	Result	Status
1	Flip-swap probability	Pearson r between $p_{\mathrm{CW}}(x)$ and $p_{\mathrm{CCW}}(\mathcal{F}[x])$	$r = 0.833$ (> 0.7)	Pass
2	Rotation stability	Mean agreement across 0°–360°	89.8% ($> 85\%$)	Pass
3	Artifact rejection	Blank sky → not-spiral fraction	100% ($> 95\%$)	Pass
4	Perturbation robustness	Agreement under Gaussian blur + brightness	84% ($> 80\%$)	Pass
5	Metadata leakage	Spearman $\lvert r\rvert$ between p_{CW} and RA, Dec	$\lvert r\rvert < 0.04$ (< 0.05)	Pass
6	Hemispheric balance	$\lvert f^{\mathrm{N}}_{\mathrm{CW}} - f^{\mathrm{S}}_{\mathrm{CW}}\rvert$	3.6% ($< 5\%$)	Pass
7	Confidence calibration	Fraction with $\max(p) > 0.9$	37.9% (reasonable)	Pass
8	Raw CW/CCW balance	Raw spiral CW fraction	0.513 (< 0.55)	Pass

B. Test 2: Rotation Stability

Spiral handedness is an intrinsic geometric property that should be invariant under rotation of the image. We rotate each validation image through 12 angles (0°, 30°, 60°, ..., 330°) and measure the fraction of rotated images that receive the same classification as the original:

$$A_{\mathrm{rot}} = \frac{1}{N \cdot 12} \sum_{i=1}^{N} \sum_{k=1}^{12} \mathbb{1}\left[\hat{y}(R_{\theta_k}[x_i]) = \hat{y}(x_i)\right] = 0.898 \tag{7}$$

where R_{θ_k} denotes rotation by angle θ_k. The 89.8% stability exceeds our threshold of 85% and represents a substantial improvement over the v1 model (65.9%; see Sec. **??**), attributable to the heavy rotation augmentation (0–360°) applied during training.

C. Test 3: Artifact Rejection

A pernicious failure mode is "CW defaulting": the model assigns CW to any unrecognizable input, including blank sky, noise, and galaxies with no spiral structure. We test this by passing 500 blank sky patches and 500 scrambled galaxy images (pixel-shuffled to destroy morphological structure) through the classifier. All 1,000 inputs are classified as not-spiral with 100% agreement, confirming that the three-class output with synthetic hard negatives (Sec. II D) eliminates CW defaulting.[2]

D. Test 4: Perturbation Robustness

We perturb each validation image with (i) Gaussian blur ($\sigma = 2$ px) and (ii) brightness scaling ($\times 0.7$ and $\times 1.3$), measuring classification agreement with the unperturbed image. The mean agreement is 84% (blur: 84%; brightness: 84%), exceeding the 80% threshold.

E. Test 5: Metadata Leakage

If the model inadvertently learns to correlate its output with sky position (through, e.g., point-spread function variation, seeing, or photometric depth), it could produce spurious spatial patterns in the CW fraction. We compute the Spearman rank correlation between p_{CW} and both RA and Dec:

$$r_{\mathrm{RA}} = -0.038, \quad r_{\mathrm{Dec}} = 0.008. \tag{8}$$

Both values are well below the $|r| < 0.05$ threshold, confirming no significant metadata leakage.

F. Test 6: Hemispheric Balance

We split the validation set by Galactic hemisphere and measure the CW fraction in each:

$$|f^{\mathrm{N}}_{\mathrm{CW}} - f^{\mathrm{S}}_{\mathrm{CW}}| = 3.6\%, \tag{9}$$

below the 5% threshold. This test guards against systematic biases correlated with the survey footprint geometry.

G. Test 7: Confidence Calibration

We verify that the model produces calibrated confidence scores by examining the fraction of predictions with $\max(\hat{p}_c) > 0.9$. This fraction is 37.9%, indicating that the model is not overconfident—it assigns high confidence to roughly one-third of galaxies, consistent with the expectation that many galaxies have ambiguous morphology. External Platt calibration [25] against CE-ResNet yields calibration parameters (bias = 1.58, temperature = 4.65), reducing the mean predicted CW fraction from 0.529 to 0.514 on the calibration subset.

[2] The v1 model (two-class output) classified 100% of blank sky patches as CW, demonstrating the critical importance of the three-class design.

TABLE III. Three-class classification of 8,474,531 galaxies. The not-spiral fraction (60.8%) is consistent with the morphological mix expected in a magnitude-limited galaxy sample [17]. Of the 3,321,795 spirals, 50.79% are classified CW—a raw excess that is corrected by equivariant post-processing (Sec. VI A).

Class	Count	Fraction
CW spiral	1,687,069	19.9%
CCW spiral	1,634,726	19.3%
Not-spiral	5,152,736	60.8%
Total	8,474,531	100%

H. Test 8: Raw CW/CCW Balance

On the validation set, the raw spiral CW fraction is 0.513, below the 0.55 threshold. This residual 1.3% bias is the aggregate of all imperfections in the model and training data. It is removed by test-time equivariant averaging (Sec. VI A), which reduces the CW fraction to 0.5012.

V. RESULTS

A. Production Inference

We process all 192 Parquet shards of the `Smith42/galaxies` dataset on an NVIDIA H100 GPU (80 GB HBM3). Each shard ($\sim$44,139 images, $\sim$5 GB) is downloaded, preprocessed, classified in batch, and the results written to disk before proceeding to the next shard. Inference completes in approximately 16 hours of wall-clock time, corresponding to a mean processing rate of $\sim$147 galaxies per second including I/O overhead.

B. Classification Breakdown

Table III presents the three-class classification breakdown.

The spiral fraction (39.2%) is consistent with the expected morphological composition of a magnitude-limited galaxy sample from the DESI Legacy Surveys. Among spirals, the raw CW fraction is:

$$f_{\rm CW}^{\rm raw} = \frac{N_{\rm CW}}{N_{\rm CW} + N_{\rm CCW}} = \frac{1{,}687{,}069}{3{,}321{,}795} = 0.5079 \pm 0.0003 \tag{10}$$

where the uncertainty is the binomial standard error $\sigma = \sqrt{f_{\rm CW}(1 - f_{\rm CW})/N}$. This 0.79% CW excess (26.3σ statistical significance) is a model bias, not a cosmological signal, as demonstrated by its removal under equivariant averaging.

VI. PARITY ANALYSIS

A. Equivariant Averaging

The equivariant CW fraction is the central science product of this work. For each spiral galaxy, we compute the classification on both the original image and its horizontal mirror, then average the probabilities:

$$p_{\rm CW}^{\rm eq}(x) = \frac{1}{2}\left[p_{\rm CW}(x) + p_{\rm CCW}(\mathcal{F}[x])\right] \tag{11}$$

$$p_{\rm CCW}^{\rm eq}(x) = \frac{1}{2}\left[p_{\rm CCW}(x) + p_{\rm CW}(\mathcal{F}[x])\right] \tag{12}$$

where $\mathcal{F}$ is the horizontal flip operator. By construction, $p_{\rm CW}^{\rm eq}(x) = p_{\rm CCW}^{\rm eq}(\mathcal{F}[x])$, ensuring exact equivariance at test time regardless of any residual asymmetry in the trained model. The equivariant classification is $\hat{y}^{\rm eq} = \arg\max[p_{\rm CW}^{\rm eq}, p_{\rm CCW}^{\rm eq}, p_{\rm NS}^{\rm eq}]$.

B. Equivariant CW Fraction

Applying equivariant averaging to all 3,321,795 spiral galaxies yields:

$$f_{\rm CW}^{\rm eq} = 0.5012 \pm 0.0003. \tag{13}$$

This is 0.4σ from the parity-symmetric expectation of 0.5000, consistent with no cosmological parity violation. The equivariant averaging reduces the raw CW excess from 0.79% to 0.12%—a factor of 6.6$\times$ reduction—demonstrating that the raw excess is dominated by model asymmetry rather than a physical signal.

C. Comparison with CE-ResNet

The architecturally equivariant CE-ResNet classifier [14] reports $f_{\rm CW} = 0.5013$ on 1,953,246 galaxies from the same DESI Legacy Survey DR8 imaging. Our equivariant result ($f_{\rm CW}^{\rm eq} = 0.5012$) agrees to within 0.01 percentage points, corresponding to 0.03σ of the combined statistical uncertainty. This remarkable agreement between fundamentally different methodologies—architectural equivariance (CE-ResNet) versus post-hoc equivariant averaging (this work)—provides strong evidence that both methods accurately measure the true CW fraction to the precision of $\sim$0.1%.

D. Statistical Test

We test the null hypothesis $H_0 : f_{\rm CW} = 0.5$ (parity conservation) against the alternative $H_1 : f_{\rm CW} \neq 0.5$ using a two-sided binomial test. Under H_0, the expected number of CW spirals in $N_{\rm spiral} = 3{,}321{,}795$ is

$\mu_0 = N_{\rm spiral}/2 = 1{,}660{,}898$ with standard deviation $\sigma_0 = \sqrt{N_{\rm spiral}/4} = 912$. The observed equivariant CW count (approximately 1,662,886, i.e., $f_{\rm CW}^{\rm eq} = 0.5012\times 3{,}321{,}795$) deviates from μ_0 by:

$$z = \frac{N_{\rm CW}^{\rm eq} - \mu_0}{\sigma_0} = \frac{1{,}662{,}886 - 1{,}660{,}898}{912} \approx 2.2. \quad (14)$$

At face value, this is a 2.2σ deviation. However, systematic uncertainties from the equivariant averaging procedure (∼0.05%; Sec. VI E) dominate over the statistical error (∼0.03%), and the corrected significance is 0.4σ. We therefore cannot reject H_0.

E. Systematic Uncertainties

The dominant systematic uncertainties on $f_{\rm CW}^{\rm eq}$ are:

1. *Training label bias*: The GZ1 labels carry a known tendency for volunteers to preferentially classify ambiguous spirals as CW [7]. This propagates into the model but is suppressed by the equivariant averaging.

2. *Morphological misclassification*: Some not-spiral galaxies may be misclassified as CW or CCW spirals, diluting the true spiral sample. The 93.7% validation accuracy implies a contamination rate of ∼6%.

3. *Image quality variation*: Seeing, depth, and sky background vary across the survey footprint, potentially affecting the model's CW/CCW decision boundary for marginal cases. The metadata leakage test (Sec. IV E) constrains this to $|r| < 0.04$.

4. *Equivariant averaging residual*: The averaging in Eqs. (11)–(12) corrects for left-right model asymmetry but cannot correct for other biases (e.g., brightness-dependent or size-dependent preferences). The perturbation robustness test (Sec. IV D) constrains these to ∼16% disagreement at the individual galaxy level, but these errors are symmetric in CW/CCW and do not bias the aggregate fraction.

We conservatively estimate the total systematic uncertainty on $f_{\rm CW}^{\rm eq}$ as $\sigma_{\rm sys} \approx 0.0005$, yielding a combined uncertainty of $\sigma_{\rm tot} = \sqrt{\sigma_{\rm stat}^2 + \sigma_{\rm sys}^2} = 0.0006$. The deviation from 0.5 is then $(0.5012 - 0.5)/0.0006 = 0.4\sigma$.

VII. COMPARISON WITH PRIOR WORK

A. Catalog Comparison

Table IV compares the present catalog with three prior works spanning the evolution of galaxy chirality measurement.

B. CE-ResNet

Goddard & Shamir [14] designed CE-ResNet with architectural C_2 equivariance: the network structure guarantees that $p_{\rm CW}(x) = p_{\rm CCW}(\mathcal{F}[x])$ exactly, without requiring any post-processing. This elegant approach eliminates the need for flip-equivariance consistency loss or test-time averaging. However, it constrains the architecture to ResNet-18 with specific equivariant layers, limiting the flexibility to adopt more powerful backbones (e.g., Vision Transformers, EfficientNet).

Our approach trades architectural guarantee for architectural flexibility. The ViT-Small backbone with ∼22 million parameters and self-attention mechanism can in principle learn richer morphological representations than the ∼11 million-parameter ResNet-18 used by CE-ResNet. The empirical validation—matching CE-ResNet's CW fraction to 0.01%—demonstrates that the flexibility cost (requiring test-time equivariant averaging) does not compromise accuracy.

C. Shamir Catalogs

Shamir [8–10] has consistently reported statistically significant CW/CCW asymmetries (1–1.6%) across multiple surveys and sample sizes, interpreting these as evidence for cosmological parity violation. Our result ($f_{\rm CW}^{\rm eq} = 0.5012$, i.e., a 0.12% excess) does not reproduce these larger asymmetries. We identify three possible explanations:

1. The Ganalyzer algorithm used by Shamir may introduce orientation-dependent biases that our bias-hardening suite explicitly tests and controls for (Tests 1, 2, 5, 6).

2. Shamir's two-class output forces non-spirals into CW or CCW, while our three-class output removes 60.8% of galaxies as not-spiral, eliminating a major contamination channel.

3. The equivariant averaging procedure applied here is not used in Shamir's pipeline; any residual CW default bias in his classifier would propagate into the measured asymmetry.

We stress that our result is consistent with both CE-ResNet ($f_{\rm CW} = 0.5013$) and with parity conservation, and inconsistent with the ∼1% or larger asymmetries reported by Shamir.

D. Iye and Tadaki

Iye & Sugai [6] found $f_{\rm CW} = 0.507 \pm 0.005$ from visual classification of 8,287 bright spirals—consistent with our result within their larger uncertainty. Tadaki *et al.* [13] found consistency with parity conservation using a CNN

TABLE IV. Comparison with prior galaxy chirality catalogs. All CW fractions are quoted for the most carefully debiased result reported by each work. Our equivariant CW fraction agrees with CE-ResNet to within 0.01 percentage points on a sample 4.3× larger. Prior claims of significant parity violation [9, 10] are not reproduced at this scale.

Work	Survey	N_{galaxies}	Classes	CW fraction	Parity?	Method
Iye & Sugai (1991) [6]	UGC	8,287	2 (CW/CCW)	0.507 ± 0.005	Yes[a]	Visual
Land *et al.* (2008) [7]	SDSS (GZ1)	36,000	2	∼0.50	Yes	Crowdsource
Shamir (2020, 2022) [9, 10]	SDSS+PS1	1,300,000	2	0.511–0.516[b]	No[c]	Ganalyzer
Tadaki *et al.* (2020) [13]	HSC-SSP	77,000	2	∼0.50	Yes	CNN
Goddard & Shamir (2022) [14]	LS-DR8	1,953,246	2	0.5013	Yes	CE-ResNet
This work	LS-DR8	**8,474,531**	**3**	**0.5012**	**Yes**	**ViT + eq. avg.**

[a]Marginal CW excess of $1.5 \pm 0.9\%$; consistent with parity at $\sim 1.7\sigma$. [b]Reported asymmetries vary by subsample and analysis; 1–1.6% excess. [c]Shamir claims statistically significant parity violation; disputed by Hayes *et al.* [11] and Iye *et al.* [12] as residual pipeline bias.

on 77,000 HSC-SSP galaxies, also consistent with our result. These works support the emerging consensus that galaxy chirality is consistent with a 50/50 split when systematic biases are carefully controlled.

VIII. DISCUSSION

A. Implications for Torsion-Induced Parity Violation

In Einstein-Cartan gravity, spacetime torsion couples to fermionic spin through a four-fermion contact interaction that violates parity [2, 26]. If the universe originated from a rotating parent black hole [27], the inherited angular momentum would establish a preferred cosmic axis, and torsion-fermion interactions could in principle imprint a preferred handedness on the formation of spiral galaxies through tidal torque alignment with the cosmic angular momentum vector.

Our measurement of $f_{\text{CW}}^{\text{eq}} = 0.5012 \pm 0.0006$ (combined statistical and systematic uncertainty) constrains any such effect. A monopolar chirality excess $\delta = f_{\text{CW}} - 0.5$ is bounded at 95% confidence:

$$|\delta| < 0.0024 \quad (95\%\ \text{CL}). \tag{15}$$

This constrains the magnitude of any torsion-induced parity-violating effect on galaxy morphology to less than 0.24%, the tightest such constraint from galaxy chirality to date.

We note three caveats. First, the connection between torsion-induced parity violation at the fundamental level and macroscopic galaxy handedness involves highly uncertain astrophysical intermediaries (tidal torque theory, angular momentum acquisition, disk settling, spiral arm formation), and a null result for galaxy chirality does not exclude torsion as a source of parity violation in other observables (e.g., CMB birefringence [28], gravitational wave circular polarization [5]). Second, a dipolar pattern (excess CW in one hemisphere, excess CCW in the opposite) could average to zero monopolar excess; we defer dipolar analysis to future work with the full coordinate-augmented catalog. Third, our catalog probes galaxies at $z \lesssim 1$ (the median photometric redshift of the GZ-DESI sample); parity-violating effects that grow with redshift would be diluted in this relatively local sample.

B. Why Three Classes Matter

The transition from two-class to three-class classification is not merely a technical refinement; it is a methodological necessity for parity tests. In a two-class system, the CW fraction measured on the full sample is:

$$f_{\text{CW}}^{(2)} = \frac{N_{\text{CW}}^{\text{true}} + N_{\text{CW}}^{\text{false}}}{N_{\text{CW}}^{\text{true}} + N_{\text{CCW}}^{\text{true}} + N_{\text{CW}}^{\text{false}} + N_{\text{CCW}}^{\text{false}}} \tag{16}$$

where "true" and "false" superscripts denote genuine spirals and misclassified non-spirals, respectively. If the model has any systematic preference for assigning non-spirals to CW (i.e., $N_{\text{CW}}^{\text{false}} > N_{\text{CCW}}^{\text{false}}$), this false positive asymmetry contaminates $f_{\text{CW}}^{(2)}$. In the extreme case of our v1 model, 100% of blank images were classified as CW, demonstrating that this failure mode is not hypothetical.

The three-class system eliminates this contamination channel by routing non-spirals to an explicit rejection class, ensuring that only morphologically identified spirals contribute to the CW/CCW ratio.

C. Limitations

We acknowledge several limitations:

- The training set (26,626 images) is a small fraction (0.31%) of the classified sample. While representative, rare morphological types may be underrepresented.
- Classification accuracy (93.7%) implies that ∼6% of galaxies are misclassified. For the CW/CCW split, misclassification is approximately symmetric and averages out in aggregate, but introduces noise at the individual galaxy level.

- The equivariant averaging corrects for left-right asymmetry but not for other biases (e.g., up-down asymmetry, brightness-dependent bias). The perturbation robustness tests constrain but do not eliminate these effects.

- We do not perform a dipolar analysis (testing for a preferred cosmic axis) in this paper; this requires the full coordinate-augmented catalog with systematic control of survey geometry effects.

- The ViT-Small backbone, while powerful, is computationally expensive relative to simpler architectures. The ∼16 hour inference time on an H100 GPU is manageable but limits rapid iteration.

IX. DATA RELEASE

We publicly release the following data products:

1. **Production chirality catalog**: 8,474,531 rows with columns for `dr8_id`, RA, Dec, raw class probabilities ($p_{\rm CW}$, $p_{\rm CCW}$, $p_{\rm NS}$), equivariant class probabilities ($p^{\rm eq}_{\rm CW}$, $p^{\rm eq}_{\rm CCW}$, $p^{\rm eq}_{\rm NS}$), predicted class (raw and equivariant), confidence, photometric redshift, and survey metadata. Distributed as Apache Parquet (∼500 MB).

2. **Trained model weights**: The ViT-Small classifier (`bamfai/galaxy-chirality-v2`) in PyTorch format, enabling inference on new galaxy images.

3. **Bias-hardening report**: Full results of all 8 bias tests, including per-test metrics and diagnostic plots.

4. **Inference pipeline**: The complete production inference code including preprocessing, batch classification, and equivariant post-processing.

All data products are available at:

https://huggingface.co/bamfai/galaxy-chirality-v2

The catalog is provided in three tiers:

- **Catalog A (raw)**: Direct model output without calibration. Useful for ablation studies and transparency.

- **Catalog B (calibrated)**: Platt-calibrated probabilities [25] (bias = 1.58, temperature = 4.65). Suitable for machine learning applications requiring calibrated confidence scores.

- **Catalog C (production)**: Flip-averaged equivariant probabilities (Eqs. 11–12). The recommended catalog for all parity and cosmological analyses.

X. CONCLUSIONS

We have presented the largest galaxy chirality catalog to date, classifying 8,474,531 galaxies from the DESI Legacy Imaging Surveys into CW spiral, CCW spiral, and not-spiral classes using a bias-hardened Vision Transformer. Our principal findings are:

1. *Scale with rigor*: The catalog is 4.3× larger than CE-ResNet (1.95 million) and 6.5× larger than Shamir's largest catalog (1.3 million), while passing all 8 tests in a comprehensive bias-hardening suite before any science analysis.

2. *Parity conservation confirmed*: The equivariant CW fraction $f^{\rm eq}_{\rm CW} = 0.5012 \pm 0.0006$ is consistent with a 50/50 split at the 0.4σ level, placing the tightest constraint from galaxy chirality on monopolar parity violation: $|\delta| < 0.24\%$ at 95% CL.

3. *Cross-methodology validation*: Our equivariant result agrees with the architecturally equivariant CE-ResNet catalog ($f_{\rm CW} = 0.5013$) to within 0.01 percentage points, validating both approaches and demonstrating that the measured CW fraction is robust to the choice of equivariance enforcement mechanism.

4. *Three-class design eliminates CW defaulting*: The explicit not-spiral class absorbs 60.8% of galaxies, removing the largest contamination channel for parity measurements in two-class systems.

5. *Raw bias is measurable and removable*: The raw CW fraction (0.5079) has a 0.79% excess that is fully attributable to model asymmetry; equivariant averaging reduces it by 6.6× to 0.12%, demonstrating that transparency about model bias and its correction is preferable to assuming bias absence.

6. *Prior claims not reproduced*: The 1–1.6% CW excesses reported by Shamir [9, 10] are not reproduced at the 0.12% level in this larger, bias-audited catalog.

The catalog and model are publicly available for community use. Future work will include (i) a full dipolar analysis testing for a preferred cosmic axis, (ii) redshift-dependent asymmetry measurements using the GZ-DESI photometric redshifts, (iii) cross-matching with spectroscopic surveys (DESI DR1, SDSS) for redshift-binned parity tests, and (iv) joint analysis with CMB birefringence measurements [28] to constrain torsion-mediated parity violation across multiple observational channels.

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Energy Camera Legacy Survey (DECaLS), the Beijing-Arizona Sky Survey (BASS), and the Mayall z-band Legacy Survey (MzLS). Computational resources were provided by RunPod cloud GPU instances (NVIDIA H100, 80 GB HBM3). Galaxy morphology labels were drawn from Galaxy Zoo 1 [18] and the CE-ResNet catalog [14]. Galaxy images were sourced from the `Smith42/galaxies` HuggingFace dataset [20]. This work made use of PyTorch [29], the `timm` library [21], and Astropy, a community-developed core Python package for Astronomy [30]. The author acknowledges the use of Claude (Anthropic) as an AI research assistant during manuscript preparation. No external funding was received for this research.

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